Step 6b: Neyman Orthogonality, the Influence Function, and Rate Conditions for Dyadic AIPW

Working notes — HSG Network Econometrics paper

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What this note does. Step 6a showed that the dyadic AIPW estimator is (doubly) robust. This note asks the next question: *when the nuisance functions are estimated, how does that affect the estimator?* We show, step by step,

1. that the AIPW kernel is *Neyman-orthogonal*: small nuisance errors move it only at second order (§2);
2. why this lets us compute the asymptotics as if the nuisances were known (§3);
3. the influence function of dyadic AIPW, via the Hájek projection (§4);
4. an exact variance comparison: AIPW is never less efficient than Hájek-IPW, with a closed-form gap (§5);
5. the rate conditions under which ML nuisances may be plugged in (§6), and what carries over to the multiply robust kernel (§7).

Every step is tied to the course (Graham, Lecture 7 / Handbook Ch. 2 Sec. 5) and to Steps 1–6a of our walkthrough.

1 Setup, notation and two workhorse lemmas

Setup (Graham's dyadic ASF model)

- Agents $i = 1, \dots, N$ are drawn i.i.d. Agent i carries $Z_i = (R_i, S_i, A_i, B_i, W_i, X_i)$: sender proxy R_i, receiver proxy S_i, latent sender type A_i, latent receiver type B_i, sender policy W_i (e.g. export subsidy), receiver policy X_i (e.g. tariff).
- Outcomes: $Y_{ij} = h(W_i, X_j, A_i, B_j, V_{ij})$, with pair shocks V_{ij} i.i.d. and independent of all Z's (A3).
- Fix the policy pair (w, x). Write $\bar{h}(a, b) := \mathbb{E}_V[h(w, x, a, b, V)]$.
- Target (dyadic ASF at (w, x)): $m := \mathbb{E}\big[\bar{h}(A_1, B_2)\big]$, for two independent agents.
- Selection on proxies (A4): $A_i \perp\!\!\!\perp W_i \mid R_i$ and $B_j \perp\!\!\!\perp X_j \mid S_j$. Overlap (A5): p_w, p_x bounded away from 0.

Nuisance functions and derived objects

$$p_w(r) = \Pr(W_i = w \mid R_i = r), \qquad p_x(s) = \Pr(X_j = x \mid S_j = s),$$
$$a_i = \frac{\mathbf{1}\{W_i = w\}}{p_w(R_i)}, \qquad b_j = \frac{\mathbf{1}\{X_j = x\}}{p_x(S_j)},$$
$$g^e(a) = \int \bar{h}(a,b)\, dF_B(b), \qquad g^a(b) = \int \bar{h}(a,b)\, dF_A(a),$$
$$q^e(r) = \int q(r,s)\, f_S(s)\, ds, \qquad q^a(s) = \int q(r,s)\, f_R(r)\, dr,$$

$$q(r,s) = \mathbb{E}[Y_{ij} \mid W_i = w, X_j = x, R_i = r, S_j = s] \qquad \text{(Graham's proxy regression)}.$$

g^e, g^a: ASF of one fixed sender / receiver type. q^e, q^a: Graham's partial averages (Lemma 1). Generic (possibly wrong) nuisance values are written with a tilde: $\tilde{q}, \tilde{p}_w, \tilde{p}_x$, with weights $\tilde{a}_i = \mathbf{1}\{W_i = w\}/\tilde{p}_w(R_i)$, $\tilde{b}_j = \mathbf{1}\{X_j = x\}/\tilde{p}_x(S_j)$, and *ratios*

$$\rho_w(r) = \frac{p_w(r)}{\tilde{p}_w(r)}, \qquad \rho_x(s) = \frac{p_x(s)}{\tilde{p}_x(s)}, \qquad u = \rho_w - 1, \quad v = \rho_x - 1, \quad \delta = q - \tilde{q}.$$

u, v, δ are the three *nuisance errors*; all three are zero at the truth.

Throughout, a tilde-nuisance is treated as a **fixed function** (not depending on the observations it is applied to). This is exactly what cross-fitting by agents guarantees (see the 18 Oct discussion, and §6).

Two lemmas we will use again and again

Lemma 1 (Weight lemma)**.** *For any function $f(A_i, R_i)$ with finite mean,*

$$\mathbb{E}\big[\tilde{a}_i\, f(A_i, R_i) \,\big|\, R_i\big] = \rho_w(R_i)\, \mathbb{E}\big[f(A_i, R_i) \,\big|\, R_i\big],$$

and in particular, at the true score ($\rho_w = 1$), $\mathbb{E}[a_i f \mid R_i] = \mathbb{E}[f \mid R_i]$ and $\mathbb{E}[a_i \mid R_i] = 1$. The same holds for $\tilde{b}_j$ with (B_j, S_j, ρ_x).

Proof. Condition on R_i, so $\tilde{p}_w(R_i)$ is a constant and can be pulled out:

$$\begin{aligned}
\mathbb{E}[\tilde{a}_i f \mid R_i] &= \frac{1}{\tilde{p}_w(R_i)}\, \mathbb{E}\big[\mathbf{1}\{W_i = w\} f(A_i, R_i) \mid R_i\big] \\
&= \frac{1}{\tilde{p}_w(R_i)}\, \Pr(W_i = w \mid R_i)\, \mathbb{E}\big[f(A_i, R_i) \mid R_i, W_i = w\big] && \text{(law of total expectation)} \\
&= \frac{p_w(R_i)}{\tilde{p}_w(R_i)}\, \mathbb{E}\big[f(A_i, R_i) \mid R_i\big] && \text{(A4: } A_i \perp\!\!\!\perp W_i \mid R_i\text{)}.
\end{aligned}$$

□

Intuition

This is the single-agent IPW argument in one line: the weight undoes the probability of being treated, and A4 says that, within a proxy cell, the treated have the same type distribution as everyone else. If the score is wrong, the weight undoes the wrong probability and leaves the factor ρ_w.

Lemma 2 (Partial averages are conditional ASFs)**.** *$q^e(r) = \mathbb{E}[g^e(A_i) \mid R_i = r]$ and $q^a(s) = \mathbb{E}[g^a(B_j) \mid S_j = s]$.*

Proof. Take distinct agents $i \neq j$. Because V_{ij} is independent of everything (A3), $q(r,s) = \mathbb{E}[\bar{h}(A_i, B_j) \mid W_i = w, X_j = x, R_i = r, S_j = s]$. Agents i and j are independent draws, so $(A_i, R_i, W_i) \perp\!\!\!\perp (B_j, S_j, X_j)$; combined with A4 on each side we may drop both policy conditions:

$$q(r,s) = \iint \bar{h}(a,b)\, dF_{A|R=r}(a)\, dF_{B|S=s}(b).$$

Integrate over s with the marginal f_S; since $\int dF_{B|S=s}(b)\, f_S(s)\, ds = dF_B(b)$,

$$q^e(r) = \int \Big[\int \bar{h}(a,b)\, dF_B(b) \Big] dF_{A|R=r}(a) = \mathbb{E}[g^e(A_i) \mid R_i = r].$$

The receiver side is symmetric. □

In words: $g^e(A_i)$ is what country i *really* exports on average in the counterfactual; $q^e(R_i)$ is the best prediction of that from its proxies. Their difference $g^e(A_i) - q^e(R_i)$ is "what the proxies miss"; by Lemma 2 it has mean zero given R_i.

2 Neyman orthogonality

2.1 The kernel and its exact conditional bias

The dyadic AIPW kernel with generic nuisances $\tilde{\eta} = (\tilde{q}, \tilde{p}_w, \tilde{p}_x)$ is

$$\phi_{ij}(\tilde{\eta}) = \tilde{q}(R_i, S_j) + \tilde{a}_i \tilde{b}_j \big(Y_{ij} - \tilde{q}(R_i, S_j)\big), \qquad \hat{m}(\tilde{\eta}) = \frac{1}{N(N-1)} \sum_{i \neq j} \phi_{ij}(\tilde{\eta}).$$

Proposition 1 (Exact bias)**.** *For any fixed nuisances $\tilde{\eta}$,*

$$\mathbb{E}\big[\phi_{ij}(\tilde{\eta}) \mid R_i, S_j\big] = \tilde{q} + \rho_w \rho_x\, (q - \tilde{q}), \qquad \textit{hence} \qquad B(\tilde{\eta}) := \mathbb{E}[\hat{m}(\tilde{\eta})] - m = \mathbb{E}\big[(u + v + uv)\, \delta\big],$$

where $u = u(R_i)$, $v = v(S_j)$, $\delta = \delta(R_i, S_j)$ and $R_i \perp\!\!\!\perp S_j$ (distinct agents).

Proof. Step 1 (the weighted outcome). Condition on (R_i, S_j). The product $\tilde{a}_i \tilde{b}_j$ is nonzero only if $W_i = w$ and $X_j = x$, so

$$\mathbb{E}[\tilde{a}_i \tilde{b}_j Y_{ij} \mid R_i, S_j] = \frac{\Pr(W_i = w, X_j = x \mid R_i, S_j)}{\tilde{p}_w(R_i) \tilde{p}_x(S_j)}\, \mathbb{E}[Y_{ij} \mid W_i = w, X_j = x, R_i, S_j].$$

Because i and j are independent agents, the joint policy probability **factorises**: $\Pr(W_i = w, X_j = x \mid R_i, S_j) = p_w(R_i)\, p_x(S_j)$. (This is the Step 1 product-score argument; it is where random sampling of agents enters.) The last expectation is $q(R_i, S_j)$ by definition. So the first piece equals $\rho_w \rho_x\, q$.

Step 2 (the weighted fitted value). $\tilde{q}(R_i, S_j)$ is fixed given (R_i, S_j), so $\mathbb{E}[\tilde{a}_i \tilde{b}_j \tilde{q} \mid R_i, S_j] = \tilde{q}\, \mathbb{E}[\tilde{a}_i \tilde{b}_j \mid R_i, S_j] = \rho_w \rho_x\, \tilde{q}$ by the same factorisation.

Step 3 (combine). $\mathbb{E}[\phi_{ij} \mid R_i, S_j] = \tilde{q} + \rho_w \rho_x (q - \tilde{q})$. Subtract q:

$$\mathbb{E}[\phi_{ij} \mid R_i, S_j] - q = (\rho_w \rho_x - 1)(q - \tilde{q}) = \big((1+u)(1+v) - 1\big)\delta = (u + v + uv)\, \delta.$$

Step 4 (unconditional). By Graham's Theorem 2, $m = \iint q(r,s) f_R(r) f_S(s)\, dr\, ds = \mathbb{E}[q(R_i, S_j)]$ for $i \neq j$. Taking expectations of Step 3 over (R_i, S_j) and averaging over pairs gives $B(\tilde{\eta})$. □

Why "product of errors" is the whole story

Every term of B contains δ (outcome-model error) *multiplied by* a score error. If either error is zero, the bias is zero (double robustness, Step 6a). If both are small, the bias is

the product of two small numbers — much smaller. Orthogonality is the formal name for this second property.

2.2 Orthogonality: the first derivative vanishes

Definition (Neyman orthogonality, informal)

Move the nuisances along a straight path from the truth η_0 in an arbitrary direction: $q_t = q + t\Delta_q$, $p_{w,t} = p_w + t\Delta_w$, $p_{x,t} = p_x + t\Delta_x$, $t \in [0,1]$. The estimator is *Neyman-orthogonal* if $\frac{d}{dt} B(\eta_t)\Big|_{t=0} = 0$ for every direction $(\Delta_q, \Delta_w, \Delta_x)$.

Proposition 2. *Dyadic AIPW is Neyman-orthogonal with respect to* (q, p_w, p_x).

Proof. Compute the three errors along the path.

$$\delta_t = q - q_t = -t\Delta_q,$$
$$u_t = \frac{p_w}{p_w + t\Delta_w} - 1 = -\frac{t\Delta_w}{p_w + t\Delta_w} = -t\frac{\Delta_w}{p_w} + O(t^2), \qquad v_t = -t\frac{\Delta_x}{p_x} + O(t^2).$$

(Overlap A5 keeps p_w, p_x away from zero, so these expansions are valid.) Plug into Proposition 1:

$$B(\eta_t) = \mathbb{E}\big[(u_t + v_t + u_t v_t)\,\delta_t\big] = \mathbb{E}\Big[\Big(-t\tfrac{\Delta_w}{p_w} - t\tfrac{\Delta_x}{p_x} + O(t^2)\Big)(-t\Delta_q)\Big] = t^2\,\mathbb{E}\Big[\Big(\tfrac{\Delta_w}{p_w} + \tfrac{\Delta_x}{p_x}\Big)\Delta_q\Big] + O(t^3).$$

$B(\eta_t)$ is quadratic in t, so its derivative at $t = 0$ is zero, for every direction. □

2.3 Contrast with the other two estimators

(a) Proxy regression (Graham's estimator): kernel $\phi_{ij}^{\mathrm{REG}} = \tilde{q}(R_i, S_j)$. Its bias is $B = \mathbb{E}[\tilde{q} - q] = t\,\mathbb{E}[\Delta_q]$: **first order** in the outcome-model error. This is why, in Graham's Lemma 2, estimating the regression parameter γ *adds* a term to the asymptotic variance.

(b) IPW : kernel $\phi_{ij}^{\mathrm{IPW}} = \tilde{a}_i \tilde{b}_j Y_{ij}$. Step 1 of Proposition 1 gives $\mathbb{E}[\phi^{\mathrm{IPW}} \mid R_i, S_j] = \rho_w \rho_x q$, so

$$B(\eta_t) = \mathbb{E}[(\rho_{w,t}\rho_{x,t} - 1)q] = -t\,\mathbb{E}\Big[\tfrac{\Delta_w(R_i)}{p_w(R_i)}\,q(R_i, S_j)\Big] - t\,\mathbb{E}\Big[\tfrac{\Delta_x(S_j)}{p_x(S_j)}\,q(R_i, S_j)\Big] + O(t^2).$$

Integrate the first term over S_j: $\mathbb{E}[\Delta_w/p_w \cdot q^e(R)]$. **First order** in the score error.

Link to Step 4. For a parametric score $p_w(r;\alpha)$, a perturbation $\tilde{\alpha} - \alpha$ gives $\Delta_w/p_w \approx d_w(R)'(\tilde{\alpha} - \alpha)$ with $d_w = \partial \log p_w / \partial \alpha$. The first-order bias is then $-\mathbb{E}[d_w(R) q^e(R)]'(\tilde{\alpha} - \alpha) = M_\alpha(\tilde{\alpha} - \alpha)$ — exactly the correction term of Step 4. So M_α *is* IPW's non-zero orthogonality derivative; for AIPW the analogous derivative is zero.

Summary of §2

Estimator	Bias when nuisances are off by t	Consequence
(a) Proxy regression	$t\,\mathbb{E}[\Delta_q]$ (first order)	Lemma 2: extra variance term from $\hat{\gamma}$
(b) IPW	$-t\,\mathbb{E}[(\Delta_w/p_w)q^e] - \ldots$ (first order)	Step 4: correction M_α
AIPW	$O(t^2)$	no correction term at all

3 From orthogonality to "as if the nuisances were known"

Let $\hat{\eta}$ be the estimated nuisances and η_0 the truth. Add and subtract:

$$\hat{m}(\hat{\eta}) - m = \underbrace{\big[\hat{m}(\eta_0) - m\big]}_{T_1\text{: oracle}} + \underbrace{\big[\hat{m}(\hat{\eta}) - \hat{m}(\eta_0) - B(\hat{\eta})\big]}_{T_2\text{: fluctuation}} + \underbrace{B(\hat{\eta})}_{T_3\text{: bias}}.$$

(We used $B(\eta_0) = 0$.) Each term in words:

- T_1 is the estimator we would compute if we knew the nuisances. It is a U-statistic; §4 shows $T_1 = \frac{1}{N}\sum_i \psi_i + o_p(N^{-1/2})$.
- T_2 is the *random* part of the effect of using $\hat{\eta}$ instead of η_0: the sample average of $\phi(\hat{\eta}) - \phi(\eta_0)$ minus its mean. With cross-fitting by agents, $\hat{\eta}$ is fixed given the training folds, so T_2 is a mean-zero average whose variance is of order $\|\hat{\eta} - \eta_0\|^2/N$. It is $o_p(N^{-1/2})$ as soon as $\hat{\eta}$ is consistent. (With parametric nuisances, no cross-fitting is needed: standard stochastic-equicontinuity arguments apply.)
- T_3 is the *systematic* part. By Proposition 1 it is *exactly* $\mathbb{E}[(u + v + uv)\delta]$ evaluated at $\hat{\eta}$: a product of errors. §6 gives conditions making it $o_p(N^{-1/2})$.

Without orthogonality, T_3 would be first order in $\hat{\eta} - \eta_0$ and would add a term to the influence function (IPW: M_α; regression: Lemma 2). With orthogonality it does not, so

$$\hat{m}(\hat{\eta}) - m = \frac{1}{N}\sum_{i=1}^{N} \psi_i + o_p(N^{-1/2}),$$

where ψ_i is computed at the **true** nuisances. That is what we do next.

4 The influence function (Hájek projection at the truth)

4.1 Projection formula (recap of Step 3)

For the U-statistic $U = \frac{1}{N(N-1)}\sum_{i\neq j}\phi_{ij}$ with kernel at the truth, the Hájek projection is $\sum_k \mathbb{E}[U - m \mid Z_k]$. Only the $2(N-1)$ dyads involving agent k contribute ($N-1$ as sender, $N-1$ as receiver); dyads not involving k are independent of Z_k. Hence

$$\mathbb{E}[U - m \mid Z_k] = \frac{(N-1)\big(\phi^e(Z_k) - m\big) + (N-1)\big(\phi^a(Z_k) - m\big)}{N(N-1)} = \frac{1}{N}\,\psi_k,$$

$$\phi^e(z) := \mathbb{E}[\phi_{kj} \mid Z_k = z] \quad (k \text{ as sender}), \qquad \phi^a(z) := \mathbb{E}[\phi_{jk} \mid Z_k = z] \quad (k \text{ as receiver}),$$

$$\psi_k := \phi^e(Z_k) + \phi^a(Z_k) - 2m.$$

4.2 Computing the sender part

Fix agent i (condition on Z_i) and average over a random partner $j \neq i$ and the pair shock.

Regression term. S_j is independent of Z_i, so $\mathbb{E}[q(R_i, S_j) \mid Z_i] = \int q(R_i, s) f_S(s)\, ds = q^e(R_i)$.

Residual term. a_i is a known number given Z_i, so it comes out: $\mathbb{E}[a_i b_j (Y_{ij} - q(R_i, S_j)) \mid Z_i] = a_i\,\mathbb{E}[b_j(Y_{ij} - q(R_i, S_j)) \mid Z_i]$. If $W_i \neq w$, $a_i = 0$ and we are done. If $W_i = w$:

- $Y_{ij} = h(w, X_j, A_i, B_j, V_{ij})$. Condition additionally on S_j and apply the weight lemma to j's side (with $\rho_x = 1$), then average out V_{ij} (A3): $\mathbb{E}[b_j Y_{ij} \mid Z_i, S_j] = \mathbb{E}[\bar{h}(A_i, B_j) \mid S_j]$, i.e. j's tariff is "forced" to x. Averaging over S_j: $\int \bar{h}(A_i, b)\, dF_B(b) = g^e(A_i)$.
- $\mathbb{E}[b_j\, q(R_i, S_j) \mid Z_i] = \mathbb{E}[q(R_i, S_j)]_{S_j} = q^e(R_i)$, since $\mathbb{E}[b_j \mid S_j] = 1$.

So the residual term is $a_i\big(g^e(A_i) - q^e(R_i)\big)$. The receiver part is symmetric.

Theorem 1 (influence function of dyadic AIPW)

Under A1–A5, nondegeneracy $\mathrm{Var}(\psi_i) > 0$, and the conditions of §6,

$$\sqrt{N}\,(\hat{m}_{\mathrm{AIPW}} - m) \xrightarrow{d} \mathcal{N}\big(0, \mathrm{Var}(\psi_i)\big),$$

$$\psi_i = \underbrace{\Big[q^e(R_i) - m + a_i\big(g^e(A_i) - q^e(R_i)\big)\Big]}_{D_i^e:\ i \text{ as sender}} + \underbrace{\Big[q^a(S_i) - m + b_i\big(g^a(B_i) - q^a(S_i)\big)\Big]}_{D_i^a:\ i \text{ as receiver}}.$$

Reading the influence function

Each role has two parts. $q^e(R_i) - m$: how much country i's *predicted* counterfactual exports differ from average — **unweighted**. $a_i(g^e - q^e)$: what the proxies *miss* about i's true type — only this residual is weighted. Compare IPW (Step 3), $a_i g^e(A_i) - m$, which weights everything. Weights multiplying only the unpredictable part is why AIPW is more stable.

Remark. (i) $\sqrt{N}$, not N: the effective sample size is the number of agents (Step 3). (ii) The remainder of the Hoeffding decomposition — pair shocks V_{ij} and sender–receiver type interactions — is $O_p(N^{-1})$. (iii) $\mathrm{Var}(\psi_i)$ includes $2\,\mathrm{Cov}(D_i^e, D_i^a)$, non-zero when an agent's two roles are dependent (e.g. $W_i = X_i$, WTO/Eurozone membership); a Fafchamps–Gubert-type estimator $\frac{1}{N}\sum_i \hat{\psi}_i^2$ captures it automatically.

5 AIPW versus Hájek-IPW: an exact variance comparison

The Hájek-IPW sender influence is $a_i(g^e(A_i) - m)$ (Step 2 / Lemma H). Add and subtract $q^e(R_i)$:

$$a_i(g^e - m) = \underbrace{\big[q^e - m + a_i(g^e - q^e)\big]}_{D_i^e\ \text{(AIPW)}} + \underbrace{(a_i - 1)(q^e - m)}_{E_i^e\ \text{(extra noise)}}$$

(Check: the right-hand side is $q^e - m + a_i g^e - a_i q^e + a_i q^e - a_i m - q^e + m = a_i g^e - a_i m$.)

Lemma 3 (Orthogonal decomposition)**.** $\mathrm{Cov}(D_i^e, E_i^e) = 0$, *and*

$$\mathrm{Var}(E_i^e) = \mathbb{E}\Big[(q^e(R_i) - m)^2\,\frac{1 - p_w(R_i)}{p_w(R_i)}\Big].$$

Proof. Both have mean zero. Condition on R_i, so $q^e(R_i) - m$ is a constant:

$$\mathbb{E}[D^e E^e \mid R_i] = (q^e - m)^2\,\underbrace{\mathbb{E}[a_i - 1 \mid R_i]}_{=0\ \text{(Lemma 1)}} + (q^e - m)\,\mathbb{E}\big[a_i(a_i - 1)(g^e(A_i) - q^e(R_i)) \mid R_i\big].$$

For the second term note $a_i^2 = \mathbf{1}\{W_i = w\}/p_w^2$ (an indicator squared is itself), so $a_i(a_i - 1) = \mathbf{1}\{W_i = w\}\,\frac{1}{p_w}\big(\frac{1}{p_w} - 1\big)$. Then

$$\begin{aligned}\mathbb{E}\big[a_i(a_i - 1)(g^e - q^e) \mid R_i\big] &= \Big(\tfrac{1}{p_w} - 1\Big)\,\mathbb{E}\big[g^e(A_i) - q^e(R_i) \mid R_i, W_i = w\big] \\ &= \Big(\tfrac{1}{p_w} - 1\Big)\big(\mathbb{E}[g^e(A_i) \mid R_i] - q^e(R_i)\big) = 0,\end{aligned}$$

using A4 to drop $W_i = w$ and Lemma 2 for the last equality. For the variance: $\mathbb{E}[(a_i - 1)^2 \mid R_i] = \mathbb{E}[a_i^2 \mid R_i] - 1 = \frac{1}{p_w} - 1 = \frac{1 - p_w}{p_w}$. □

Proposition 3 (efficiency gain of AIPW over Hájek-IPW)

Per role,

$$\mathrm{Var}\left(a_i(g^e - m)\right) = \mathrm{Var}(D_i^e) + \mathbb{E}\Big[(q^e(R_i) - m)^2 \, \frac{1 - p_w(R_i)}{p_w(R_i)}\Big] \;\geq\; \mathrm{Var}(D_i^e),$$

and symmetrically for the receiver role. If an agent's sender and receiver sides are independent, the totals add up: $\mathrm{Var}(\psi_i^H) = \mathrm{Var}(\psi_i^{\mathrm{AIPW}}) + \mathrm{gap}^e + \mathrm{gap}^a$. (Otherwise cross-role covariances enter and the comparison holds role by role.)

When is the gap large, when is it zero?

The gap is (squared) *predictable heterogeneity* $(q^e - m)^2$ times *weight noise* $(1 - p_w)/p_w$.

- Large when proxies strongly predict counterfactual trade (e.g. GDP predicts exports) *and* treatment is rare in some proxy cells.
- Zero when $q^e(R) = m$ for all r: the proxies carry no information about the outcome, so the regression has nothing to add — but then they also are not needed for confounding control on that side.

Payoff: the Step 4.5 claim, now proven

$a_i - 1 = \big(\mathbf{1}\{W_i = w\} - p_w(R_i)\big)/p_w(R_i)$, so $E_i^e = \big(\mathbf{1}\{W_i = w\} - p_w\big) \cdot \frac{q^e(R_i) - m}{p_w(R_i)}$ has the form "(treatment residual) × (function of R)". The logit score has the same form, $s_\alpha = (\mathbf{1}\{W_i = w\} - p_w) \cdot c(R_i)$. One checks exactly as in Lemma 3 that D_i^e is uncorrelated with every such term. Therefore regressing the Hájek-IPW influence on the score functions (Step 4) removes the projection of E_i^e and leaves D_i^e untouched. With a **saturated** score every function of R is in the span, E_i^e is removed entirely, and *estimated-score Hájek-IPW attains the AIPW variance.* With a parametric logit only part of E_i^e is removed, so AIPW is strictly better whenever $(q^e - m)/p_w$ is not in the span of the score.

Numerical check

Gravity-type DGP from the 18 Oct simulation, true nuisances, $N = 400$ agents, 4000 replications (script `claude/sims/check_6b_variance.py`):

	simulated	theory
$N \cdot \mathrm{Var}(\hat{m}_{\mathrm{AIPW}})$ vs. $\mathrm{Var}(\psi^{\mathrm{AIPW}})$	7.62	7.76
$N \cdot [\mathrm{Var}(\hat{m}_H) - \mathrm{Var}(\hat{m}_{\mathrm{AIPW}})]$ vs. gap formula	3.04	2.84

Both agree up to Monte Carlo error and finite-N terms of a few percent.

6 Rate conditions for machine-learned nuisances

With ML, $\hat{\eta}$ converges slower than $N^{-1/2}$, so $T_3 = B(\hat{\eta})$ must be controlled through its product structure.

Bounding u and v. If the estimated scores respect overlap, $\hat{p}_w, \hat{p}_x \geq \kappa > 0$, then $|u| = |p_w - \hat{p}_w|/\hat{p}_w \leq |p_w - \hat{p}_w|/\kappa$, and $|u| \leq 1/\kappa$ (since $|p_w - \hat{p}_w| \leq 1$). Same for v.

Cauchy–Schwarz. With $\|\cdot\|$ the L^2 norm under the **product** distribution $F_R \otimes F_S$ (independent sender and receiver proxies):

$$|T_3| \leq |\mathbb{E}[u\delta]| + |\mathbb{E}[v\delta]| + |\mathbb{E}[uv\delta]| \leq \frac{1}{\kappa}\|\hat{p}_w - p_w\| \, \|\hat{q} - q\| + \frac{1}{\kappa}\|\hat{p}_x - p_x\| \, \|\hat{q} - q\| + \frac{1}{\kappa^2}\|\hat{p}_x - p_x\| \, \|\hat{q} - q\|.$$

(The $uv\delta$ term is third order; bounding one $|u| \leq 1/\kappa$ makes it the same order as the others.)

Rate conditions (R)

$$\|\hat{p}_w - p_w\| \cdot \|\hat{q} - q\| = o_p(N^{-1/2}), \qquad \|\hat{p}_x - p_x\| \cdot \|\hat{q} - q\| = o_p(N^{-1/2}), \qquad \hat{p}_w, \hat{p}_x \geq \kappa > 0.$$

Sufficient: every nuisance converges faster than $N^{-1/4}$, since $N^{-1/4} \cdot N^{-1/4} = N^{-1/2}$. Together with agent-level cross-fitting (for T_2), (R) delivers Theorem 1 with ML nuisances — this is dyadic double/debiased machine learning.

Three dyadic points worth a sentence each in the paper.

1. N **counts agents, not dyads.** $\hat{q}$ is fitted on $\approx N^2$ dyads, but they share agents; the information and hence the attainable rate are governed by N. Do not claim N^2-based rates.
2. **Product measure.** The norm is under $F_R \otimes F_S$: $\hat{q}$ must be accurate for *all* (sender, receiver) proxy combinations, including those rarely observed jointly treated. This is where overlap bites for the outcome model.
3. **Parametric nuisances** (logit, Poisson PVR): errors are $O_p(N^{-1/2})$, products $O_p(N^{-1})$, (R) holds trivially and cross-fitting is unnecessary — but orthogonality still removes the M_α / Lemma-2 correction terms from the standard errors.

7 What carries over to the multiply robust kernel

For the MR kernel (notes of 18 Oct, `claude/multiply_robust_aipw.md`) the exact bias is

$$B = \mathbb{E}[u\,\delta^e] + \mathbb{E}[v\,\delta^a] + \mathbb{E}[uv\,\delta], \qquad \delta^e = q^e - \hat{q}^e, \quad \delta^a = q^a - \hat{q}^a.$$

Every term is again a product of errors, so the same path argument as Proposition 2 shows orthogonality in all five nuisances (q, p_w, p_x, q^e, q^a), and the 18 Oct projection computation shows its influence function equals ψ_i^{AIPW}. Rate conditions become $\|\hat{p}_w - p_w\|\,\|\hat{q}^e - q^e\| = o_p(N^{-1/2})$ etc.

Open: $\hat{q}^e, \hat{q}^a$ are regressions on pseudo-outcomes built from $\hat{q}$ and $\hat{p}$. Their errors inherit terms like $\mathbb{E}_R[u\,\delta](s)$ — themselves products, so the argument should go through, but it needs its own lemma. This is the main remaining theory gap.

8 Where each assumption enters

Assumption	Used in
Random sampling of agents (A1)	product factorisation of policy probabilities (Prop. 1); $R_i \perp\!\!\!\perp S_j$ and Theorem 2 (Prop. 1, Step 4); counting in the projection (§4)
Pair shocks independent (A3)	averaging out V_{ij} (Lemma 2, §4); remainder is $O_p(N^{-1})$
Selection on proxies (A4)	weight lemma; Lemma 2; orthogonal decomposition (Lemma 3)
Overlap (A5), estimated overlap	expansions in Prop. 2; bounds on u, v (§6)
Nondegeneracy $\text{Var}(\psi) > 0$	$\sqrt{N}$-rate and normality (Theorem 1)
Cross-fitting by agents	$\hat{\eta}$ fixed functions; control of T_2 (§3)

Defense version (one paragraph)

The conditional bias of the dyadic AIPW kernel is $(\rho_w\rho_x - 1)(q - \tilde{q})$, so perturbing the nuisances changes the estimand only at second order: Neyman orthogonality. Hence first-step estimation does not affect the influence function, which per role is $q^e(R) - m + a(g^e - q^e)$. It differs from Hájek-IPW's by the orthogonal term $(a-1)(q^e - m)$, whose variance $\mathbb{E}[(q^e - m)^2(1 - p_w)/p_w]$ is the efficiency gain. With ML nuisances and agent-level cross-fitting, products of nuisance errors of order $o_p(N^{-1/2})$ — N the number of agents — suffice.

References (to verify before citing)
Graham (2020), *Network data*, Handbook of Econometrics 7A, Sec. 5. Hirano, Imbens & Ridder (2003), Econometrica. Chernozhukov et al. (2018), Econometrics Journal (DML). Chiang, Kato, Ma & Sasaki (2022), JBES (multiway cluster-robust DML). Graham, Imbens & Ridder, average match function (efficiency bounds).